RDSA: Resilient Distributed Safety Architecture

Defending VLMs Against Adversarial Safety Attacks via Distributed, Entangled Safety Representations

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Milestone Report — March 2026

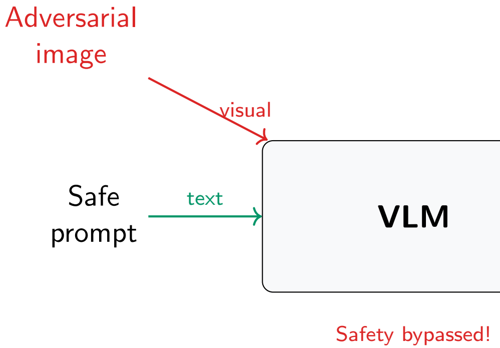
Outline

1. **Problem**: VLM safety is fragile — one attack breaks it
2. **Root cause**: Safety features are localized & separable
3. **Our approach**: RDSA — distribute, entangle, harden
4. **Technical details**: Three defense modules + theory
5. **Progress**: Implementation & preliminary results
6. **Timeline**: Path to NeurIPS 2026

VLM Safety Is Surprisingly Fragile

- VLMs (GPT-4o, Qwen-VL, Gemma) have safety alignment
- One adversarial image can bypass all safety guardrails
- SCIA attack (ICML 2026): **¿80% ASR** on aligned models
- Safety fine-tuning, RLHF — all broken by visual attacks

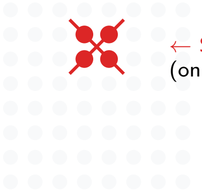
Why? → Next slide



Root Cause: Localized & Separable Safety Features

SCIA's key discovery:

- Safety behavior depends on $\sim$**0.1–0.5%** of neurons
- These neurons are disentangled from semantics
- Attacker can suppress safety **without hurting capability**



4096 neurons per layer

Analogy: Safety is a single fuse —
cut it, lights stay on, alarm goes off

Key Insight: Distribute, Entangle, Harden

Make safety features impossible to remove without destroying the model

Multi-layer redundancy

Distribute safety across layer groups

Semantic entangle

Entangle safety with semantics

Subspace AT

Harden via adversarial training

RDSA: Three Defense Modules

Theorem 1: Coverage

Theorem 2: Redundancy

Step 1: Identify
V_s, V_t via SVD/PCA

→

Step 2: Train
SA-AT + Entangle

→

Step 3: Monitor
Inference anomaly

SVD on $\Delta H = H_{\text{unsafe}} - H_{\text{safe}}$
$\rightarrow$ Safety subspace V_s [d, 32]

$\mathcal{L} = \mathcal{L}_{\text{SFT}} + \alpha_1 \mathcal{L}_{\text{SA-AT}} + \alpha_2 \mathcal{L}_{\text{consist}} + \alpha_3 \mathcal{L}_{\text{entangle}}$
LoRA fine-tuning (attention + MLP)

Cross-layer variance detection
$\text{Var}_k[\sigma(w_k^T V_s^T h_k)] > \tau$

Module 1: Subspace-Constrained Adversarial Training

Theorem 1 (SA-AT Coverage Guarantee):

- **Inner loop**: PGD finds worst-case δ^* in V_s subspace

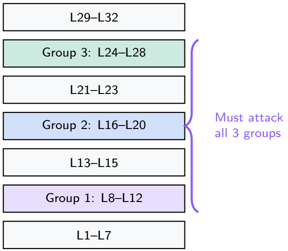
$$\delta^* = \arg\max_{\|\delta\| \leq \varepsilon} \mathcal{L}_{\mathrm{CE}}\big(f(h + V_s\delta),\ y_{\mathrm{refusal}}\big)$$

- **Outer loop**: Train model to refuse under δ^*
- Search space is only d_s**=32 dimensions** (not 4096)
- Random restarts (3×) for stronger adversarial search
- Relative ε: scaled by activation norm per layer

Module 2: Cross-Layer Safety Redundancy

Theorem 2: Attack cost amplification via G independent layer groups

- Safety distributed across G**=3 groups**
- Attack cost: $O(\varepsilon_1^*) \to O\left(\sqrt{\sum_k (\varepsilon_k^*)^2}\right)$
- Attacker must fool **all groups simultaneously**
- Non-aligned gradients across groups

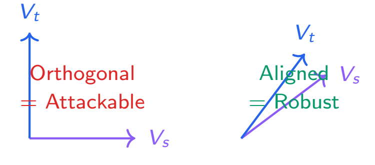


Module 3: Safety-Semantic Entanglement

Goal: Make safety inseparable from semantics

- Entanglement degree: $\eta = \frac{1}{d_s} \sum_i \max_j \left| v_s^{(i)\top} v_t^{(j)} \right|$
- $\eta \approx 0 \rightarrow$ safety removable (vulnerable)
- $\eta \approx 1 \rightarrow$ safety $=$ semantics (robust)
- **New**: $\mathcal{L}_{\text{entangle}} = 1 - \eta$
- Directly optimized via gradient descent

Attacker's dilemma: removing safety destroys model capability



$\eta \approx 0.2$ (vanilla) $\eta \approx 0.8$ (RDSA)

Training Pipeline: 10 Key Enhancements

Data & Format

- ✓ Chat template formatting
- ✓ Label masking (response only)
- ✓ LoRA: attention + MLP modules

SA-AT Improvements

- ✓ PGD random restarts ($3\times$)
- ✓ Relative ε (5% of $\|h\|$)
- ✓ Warmup: 1 epoch pure SFT first

New Losses

- ✓ $\mathcal{L}_{\text{entangle}}$: optimize η directly
- ✓ $\mathcal{L}_{\text{consist}}$: harmful samples only

Robustness

- ✓ Multi-layer hooks (all group layers)
- ✓ Intra-epoch subspace refresh

Models & Evaluation Setup

Surrogate Models (white-box)

- Qwen3-VL-8B ✓
- InternVL2.5-8B ✓
- MiniCPM-V-2.6 ✓

6 Attack Types

- SCIA, UMK (white-box)
- FigStep (visual injection)
- 3 adaptive attacks (knows RDSA)

Metrics

- ASR$\downarrow$ — Attack Success Rate
- RR$\uparrow$ — Refusal Rate
- OR$\downarrow$ — Over-Refusal Rate
- VQAv2, MMBench, MME (capability)

Safety Judge

- GPT-4o as judge (OpenAI API)
- 3 seeds for significance

Preliminary Results: Qwen3-VL-8B

Subspace Identification

- V_s: [4096, 32], V_t: [4096, 256]
- Vanilla $\eta \approx 0.49$–0.53
- 3 layer groups identified
- Singular values: $333 \to 7478$ (deep layers)

Baseline (no attack)

- Vanilla ASR $\approx$ **21.6%**
- Vanilla RR $\approx$ **78.4%**

Training Status

- SA-AT warmup epoch: running
- Consistency loss: $0.63 \to 0.02$
- Entanglement loss: $0.96 \to 0.92$
- SFT loss: converging

Expected After Training

- ASR $< 10\%$ under SCIA
- ASR $< 20\%$ under adaptive
- VQA drop $< 2\%$

NeurIPS Experiment Plan: 10 Experiments

#	Experiment	Output	Status
1	Main comparison (7 def $\times$ 6 atk $\times$ 3 models)	Table 1	In progress
2	Ablation (6 variants)	Table 2	Ready
3	SA-AT analysis (PGD, ε, restarts)	Fig 2–3	Ready
4	Cross-layer redundancy (G=1,2,3)	Fig 4–5	Ready
5	Entanglement (η profile, η vs ASR)	Fig 6	Ready
6	Capability (VQAv2, MMBench, MME)	Table 4	Ready
7	Adaptive attacks (strong budget)	Table 5	Ready
8	Transfer attack (surrogate $\rightarrow$ victim)	Table 6	Planned
9	Parameter sensitivity (7 params)	Fig 7	Ready
10	Pareto frontier (safety vs utility)	Fig 1	Planned

Total: $\sim$600 GPU-hours (1 seed) / $\sim$1800 GPU-hours (3 seeds)

Implementation Status

Completed

- ✓ Subspace identification (SVD+PCA)
- ✓ SA-AT training loop
- ✓ 4 loss functions
- ✓ 6 attack implementations
- ✓ GPT-4o safety judge
- ✓ 4 capability benchmarks
- ✓ 10 experiment scripts
- ✓ 3 open-access VLMs

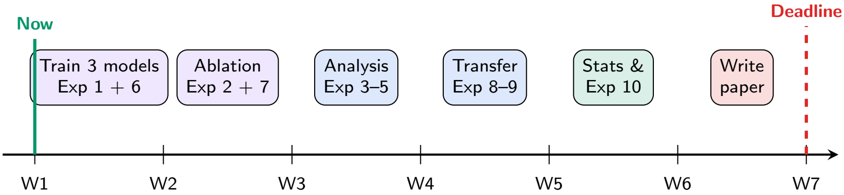
In Progress

- ○ Full RDSA training (5 epochs)
- ○ Evaluation with real attacks
- ○ Baseline defense reproduction

Codebase

- ∼5000 lines of Python
- Hydra config management
- wandb logging
- Full test suite

Timeline to NeurIPS 2026



Key risks: GPU availability, baseline reproduction, adaptive attack strength

Summary

1. **Problem**: VLM safety is fragile due to localized features
2. **RDSA**: Three-module defense (SA-AT + Redundancy + Entanglement)
3. **Theory**: Provable coverage (Thm 1) + cost amplification (Thm 2)
4. **Status**: Full implementation complete, training in progress
5. **Target**: NeurIPS 2026 — 10 experiments across 3 models

Questions?

Code: `github.com/Skyyyy0920/RDSA`